

The EUMETSAT Network of Satellite Application Facilities  NWP SAF Numerical Weather Prediction	Cloud and Aerosol Detection Software Version 3	Doc ID : NWPSAF-EC-UD-016 Version : v1.3 Date : 15/02/2024
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Cloud and Aerosol Detection Software Version 3

Release Note

version 1.3
15/02/2024

Chris Burrows
ECMWF

This documentation was developed within the context of the EUMETSAT Satellite Application Facility on Numerical Weather Prediction (NWP SAF), under the Cooperation Agreement dated 7 December 2016, between EUMETSAT and the Met Office, UK, by one or more partners within the NWP SAF. The partners in the NWP SAF are the Met Office, ECMWF, DWD and Météo France.

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Change record			
Version	Date	Author / changed by	Remarks
1.0	27/04/2020	Reima Eresmaa	Initial version
1.1	02/06/2020	Reima Eresmaa	Clarify based on beta testers' reports
1.2	02/07/2020	Reima Eresmaa	Clarify on the lack of aerosol type classification over land and that no external libraries are needed
1.3	15/02/2024	Chris Burrows	Update for CADS V3.2

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1 CLOUD AND AEROSOL DETECTION SOFTWARE VERSION V3.1

Cloud and Aerosol Detection Software (CADS) is a stand-alone detection software for cloud and aerosol contamination in satellite-based infrared (IR) radiance observations. The software is intended for use in retrieval applications and in radiance assimilation studies in the context of Numerical Weather Prediction.

Version 3.1 introduces the following new or upgraded functionalities:

- Aerosol type classification. Depending on the properties of observed IR brightness temperature, the detected aerosol is classified as either volcanic ash, Saharan dust, or other aerosol. However, the classification is currently in place for data over sea surface only.
- Separation of aerosol-affected channels from those unaffected depending on the aerosol type over sea surfaces (over land, no separation is made between affected and unaffected channels)
- Flagging for contamination by excessive trace gas concentration. In this first implementation, only detection of excessive Hydrogen Cyanide (HCN) is supported. The method can be extended to other trace gases as necessary in future releases.
- Identification of channels that are sensitive to underlying land surface.
- Extending the processing capability to radiance data from the Russian hyperspectral sounder IKFS-2 (Infrared Fourier Spectrometer -2).

As compared with the previous software version (V2.4), these changes result in an approximate 10% increase in processing time.

Users should be aware that the default settings of aerosol and trace gas detection for HIRAS, GIIRS, IKFS-2, IASI-NG, and IRS are inactive and produce no positive detections. After collecting sufficient experience from real measurement data, these settings will be revised later, and robust tuning parameter values will subsequently be provided in a future release of CADS.

2 CLOUD AND AEROSOL DETECTION SOFTWARE VERSION V3.2

CADS V3.2 has a number of minor updates:

- Support is provided for HIRAS-2.
- Support is provided for IRIS.
- Support is provided for VIIRS imager clustering data which can augment the cloud detection for CrIS.
- Bug fix to avoid potential memory leaks in the aerosol detection code.

3 INSTALLING AND RUNNING CADS

CADS is distributed in the compressed tar file CADS_V3.2.tar.gz. The file package contains the documentation (directory CADS/docs/), sample namelist files for each

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supported IR sounder (CADS/namelist/), source code files (CADS/src/), and data files and an associated test shell script (CADS/Test_Runs/).

CADS is installed by extracting the compressed tar file in a dedicated directory in the user's local file system. The installation does not require any external libraries. A quick start to verify the correctness of the installation is done as follows:

1. Go to the directory that contains all the source code files and the makefile

```
cd CADS/src
```

2. Edit the file `makefile` as necessary to comply with the user's local computing system and Fortran compiler (example settings for `gfortran`, `ifort` and `nvfortran` are provided by default)

3. Compile the software by running the command

```
make
```

in the source code directory (CADS/src/)

4. Executable CADS will appear in the parent directory CADS.
5. Go to the test run directory at

```
cd ../Test_Runs/
```

and execute the provided test script

```
./CADS_Test_Script.sh [SENSOR]
```

with [SENSOR] replaced by either AIRS, CRIS, GIIRS, HIRAS, HIRAS2, IASI, IASING, IKFS2, IRS or IRIS. The test script first makes the executable, then runs the software with input corresponding to the user's choice for [SENSOR], and at the end compares the produced output with a reference output file.

6. In the case of a successful test, the following message will be prompted on the screen:

```
Output matches with the reference.
The test run using [SENSOR] data is SUCCESSFUL.
```

7. In the case that the test run is unsuccessful, an error message of either,

```
No output produced.
The test run using the [SENSOR] data is UNSUCCESSFUL.
```

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or

Files `cads_output.dat` and `reference_output.dat` differ.
The test run using [SENSOR] data is UNSUCCESSFUL.

will be prompted on the screen. In the former case, the reason for not getting any output is likely prompted on the screen just before this error message. In the latter case it is advised that the output file (`cads_output.dat`) and the reference output file (`reference_output.dat`) are compared in detail to understand the nature of the failure.

In addition to verifying that the output files are identical to the reference output files, the scripts print out the CPU time spent while processing the input data. When running the test script on an Atos GPIL login node (AMD EPYC Rome, 2.5GHz) at ECMWF and using `gfortran`, the elapsed CPU times per spectrum, are approximately:

Instrument	Timing (s)
AIRS	0.000959
IASI	0.001528
CRIS	0.001177
GIIRS	0.000189
HIRAS	0.000938
HIRAS2	0.008
IRS	0.000471
IASING	0.001134
IRIS	0.003

4 LISTING OF FILES INCLUDED IN THE SOFTWARE DISTRIBUTION

The full listing of this package is as follows (directories typed in **bold face**):

CADS/

CADS/Test_Runs/

CADS/Test_Runs/CADS_Test_Script.sh

CADS/Test_Runs/Test_Data/

CADS/Test_Runs/Test_Data/AIRS.input.gz

CADS/Test_Runs/Test_Data/AIRS.output.gz

CADS/Test_Runs/Test_Data/CRIS.input.gz

CADS/Test_Runs/Test_Data/CRIS.output.gz

CADS/Test_Runs/Test_Data/GIIRS.input.gz

CADS/Test_Runs/Test_Data/GIIRS.output.gz

CADS/Test_Runs/Test_Data/HIRAS.input.gz

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CADS/Test_Runs/Test_Data/HIRAS.output.gz
CADS/Test_Runs/Test_Data/HIRAS2.input.gz
CADS/Test_Runs/Test_Data/HIRAS2.output.gz
CADS/Test_Runs/Test_Data/IASI.input.gz
CADS/Test_Runs/Test_Data/IASI.output.gz
CADS/Test_Runs/Test_Data/IASING.input.gz
CADS/Test_Runs/Test_Data/IASING.output.gz
CADS/Test_Runs/Test_Data/IKFS2.input.gz
CADS/Test_Runs/Test_Data/IKFS2.output.gz
CADS/Test_Runs/Test_Data/IRS.input.gz
CADS/Test_Runs/Test_Data/IRS.output.gz
CADS/Test_Runs/Test_Data/IRIS.input.gz
CADS/Test_Runs/Test_Data/IRIS.output.gz

CADS/docs/

CADS/docs/Product_Specification.pdf
CADS/docs/Release_Note.pdf
CADS/docs/Test_Plan.pdf
CADS/docs/Top_Level_Design.pdf
CADS/docs/User_Guide.pdf
CADS/docs/eresmaa_et al_itsc22_ir_screening.pdf

CADS/namelist/

CADS/namelist/AIRS_AERDET.NL
CADS/namelist/AIRS_CLDDET.NL
CADS/namelist/AIRS_LANDSENS.NL
CADS/namelist/AIRS_TRGASDET.NL
CADS/namelist/CRIS_AERDET.NL
CADS/namelist/CRIS_CLDDET.NL
CADS/namelist/CRIS_LANDSENS.NL
CADS/namelist/CRIS_TRGASDET.NL
CADS/namelist/GIIRS_AERDET.NL
CADS/namelist/GIIRS_CLDDET.NL
CADS/namelist/GIIRS_LANDSENS.NL
CADS/namelist/GIIRS_TRGASDET.NL
CADS/namelist/HIRAS_AERDET.NL
CADS/namelist/HIRAS_CLDDET.NL
CADS/namelist/HIRAS_LANDSENS.NL
CADS/namelist/HIRAS_TRGASDET.NL
CADS/namelist/HIRAS2_AERDET.NL
CADS/namelist/HIRAS2_CLDDET.NL
CADS/namelist/HIRAS2_LANDSENS.NL
CADS/namelist/HIRAS2_TRGASDET.NL
CADS/namelist/IASING_AERDET.NL
CADS/namelist/IASING_CLDDET.NL
CADS/namelist/IASING_LANDSENS.NL
CADS/namelist/IASING_TRGASDET.NL

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CADS/namelist/IASI_AERDET.NL
CADS/namelist/IASI_CLDDET.NL
CADS/namelist/IASI_LANDSENS.NL
CADS/namelist/IASI_TRGASDET.NL
CADS/namelist/IKFS2_AERDET.NL
CADS/namelist/IKFS2_CLDDET.NL
CADS/namelist/IKFS2_LANDSENS.NL
CADS/namelist/IKFS2_TRGASDET.NL
CADS/namelist/IRS_AERDET.NL
CADS/namelist/IRS_CLDDET.NL
CADS/namelist/IRS_LANDSENS.NL
CADS/namelist/IRS_TRGASDET.NL
CADS/namelist/IRIS_AERDET.NL
CADS/namelist/IRIS_CLDDET.NL
CADS/namelist/IRIS_LANDSENS.NL
CADS/namelist/IRIS_TRGASDET.NL

CADS/src/

CADS/src/CADS_Abort.F90
CADS/src/CADS_Abort.intfb
CADS/src/CADS_Detect_Aerosol.F90
CADS/src/CADS_Detect_Aerosol.intfb
CADS/src/CADS_Detect_Cloud.F90
CADS/src/CADS_Detect_Cloud.intfb
CADS/src/CADS_Detect_Cloud_Heapsort.F90
CADS/src/CADS_Detect_Cloud_Heapsort.intfb
CADS/src/CADS_Detect_Cloud_Imager.F90
CADS/src/CADS_Detect_Cloud_Imager.intfb
CADS/src/CADS_Detect_Cloud_Scenario.F90
CADS/src/CADS_Detect_Cloud_Scenario.intfb
CADS/src/CADS_Detect_Cloud_Separator.F90
CADS/src/CADS_Detect_Cloud_Separator.intfb
CADS/src/CADS_Detect_Cloud_Smooth.F90
CADS/src/CADS_Detect_Cloud_Smooth.intfb
CADS/src/CADS_Detect_Land_Sensitivity.F90
CADS/src/CADS_Detect_Land_Sensitivity.intfb
CADS/src/CADS_Detect_Trace_Gas.F90
CADS/src/CADS_Detect_Trace_Gas.intfb
CADS/src/CADS_Main.F90
CADS/src/CADS_Module.F90
CADS/src/CADS_Setup_Aerosol.F90
CADS/src/CADS_Setup_Aerosol.intfb
CADS/src/CADS_Setup_Cloud.F90
CADS/src/CADS_Setup_Cloud.intfb
CADS/src/CADS_Setup_Land_Sensitivity.F90
CADS/src/CADS_Setup_Land_Sensitivity.intfb

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CADS/src/CADS_Setup_Trace_Gas.F90
CADS/src/CADS_Setup_Trace_Gas.intfb
CADS/src/makefile